ZEERBOX · TRACEABILITY MATRIX

ZeerBox traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/zeerbox		
Source	docs/03b-traceability.md @ 44ce24e		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

ZeerBox traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 12 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-002	PR-001	Storage capacity and reach	ZBX-CAL-001, ZBX-BLD-001	Analysis of the model (CAL-001 s2), TRL 3; loading trial, TRL 4	Met on paper: 24 crates, 480 kg; top shelf 1.40 m
UR-003	PR-002	Cooling at the design point	ZBX-CAL-001, ZBX-BLD-001, ZBX-STM-001, ZBX-OPT-001	Psychrometric and heat balance (CAL-001 s4), TRL 3; logged field data, TRL 4	Met on paper: 10.7 K (9.8 to 11.4 K); store 24.3 °C
UR-004	PR-003	Humidity for produce	ZBX-CAL-001, ZBX-BLD-001, ZBX-OPT-001	Calculation (CAL-001 s5), TRL 3; logged field data, TRL 4	At risk: 81 % (77 to 85 %)
UR-005	PR-004	Humid-weather detection and mode lamp	ZBX-CAL-001, ZBX-BLD-001, ZBX-OPT-001	Logic review (CAL-001 s6), TRL 3; bench test with sensors, TRL 4	Met by logic review: 30 °C and 75 % RH gives 3.7 K and switches to hold; sensing error 0.5 K (STM S32)
UR-006	PR-005	Mold and air-quality guard	ZBX-CAL-001, ZBX-BLD-001	Logic review (CAL-001 s6), TRL 3; bench test, TRL 4	Met by logic review: hold mode about 8 air changes an hour at about 2 W; with no power for more than about 40 h the fans stop (STM S36, ZBX-DEC-001 D3)
UR-007	PR-006	Shelf life	ZBX-CAL-001	Literature comparison, TRL 3; paired storage trial with the partner, TRL 4	To verify at TRL 4: ZECC data (about 1.8 times) come from a cooler, more humid chamber
UR-008	PR-007	Solar operation	ZBX-CAL-001, ZBX-BLD-001	Energy balance (CAL-001 s7), TRL 3	Met on paper: 525 Wh from the 150 W panel (41 % margin)
UR-009	PR-008	Water use and sump	ZBX-CAL-001, ZBX-BLD-001, ZBX-STM-001	Water balance (CAL-001 s8), TRL 3	Met on paper: about 43 L (no cavity water; 11 L bleed, ZBX-DEC-001 D5); 60 L sump for 43 L
UR-010	PR-009	Low voltage and safe	ZBX-CAL-001, ZBX-BLD-001, ZBX-STM-001	Design review (CAL-001 s9), TRL 3	Met by design review: 15 A battery fuse on 2.5 mm² leads, 5 A load fuse on the 1.5 mm² load circuit, both DC rated (ZBX-DEC-001 D1, D5)
UR-011, UR-012	PR-010	Local build and repair	ZBX-CAL-001	Design review, TRL 3; parts survey and swap trials with the partner, TRL 4	To verify at TRL 4
UR-013	PR-011	Durability	ZBX-CAL-001, ZBX-BLD-001	Material review and structural checks (CAL-001 s10), TRL 3;	To verify at TRL 4; roof footings 1.8 times the uplift at

User requirement	Product requirement	Title	Evidence	Verification	Status
				supplier data and field life, TRL 4	30 m/s; beams and purlins sized
UR-014	PR-012	Affordable kit	ZBX-CAL-001, ZBX-BLD-001	Priced BOM (`bom/bom.csv`), CAL-001 s11, TRL 3	At the target: kit USD 300; structure USD 574, reported separately

Gaps

None.